

Lab 1: Exponents and Scientific Notation

The study of astronomy deals with a huge range of scales, from the properties of a single asteroid to galaxies containing trillions of stars, from sizes smaller than an atom to the edge of the Universe itself. In this lab you will practice using exponents and scientific notation to understand these extremes.

Learning Objectives

At the completion of this lab, you should be able to:

- express a number in scientific notation
- convert scientific notation to words
- multiply and divide exponents.
- do arithmetic with numbers in scientific notation.

Exponents

The study of the Universe requires both enormous and miniscule numbers to describe distance, mass, density, velocity, and other quantities. The use of exponents and powers of 10 will be important in the successful completion of labs for this course.

A positive exponent tells you how many zeros there are to the left of the decimal point, or how many zeros follow the “1”, for example:

$$10^2 = 100 \text{ or “one hundred.”}$$

$$10^9 = 1,000,000,000 \text{ or “one billion.”}$$

$$10^{43} = 10,000,000,000,000,000,000,000,000,000,000,000,000,000$$

A negative exponent tells you how many zeros precede the “1” , including one to the left of the decimal point and the rest to the right of the decimal point. For example:

$$10^{-1} = 0.1 \text{ or “one tenth.”}$$

$$10^{-3} = 0.001 \text{ or “one thousandth.”}$$

$$10^{-12} = 0.000000000001$$

A negative sign in the exponent is the same as that power being in the *denominator* (bottom of a fraction) instead of the *numerator* (top of a fraction) without the negative sign. For example, $10^{-3} = \frac{1}{10^3}$.

Convert the following numbers to exponents or decimals:

$$10^4 =$$

$$10^{-2} =$$

$$10^{-6} =$$

$$0.0001 =$$

$$100,000 =$$

$$10 =$$

Common names and prefixes for powers of 10 abound in astronomy. Complete the table below. For example, a billionth is “nano-” or 10^{-9} and is used in *nanometer* and *nanosecond*.

	Prefix	Power of 10	Example
Thousand			
Million			
Billion			
Tenth			
Hundredth			
Thousandth			
Millionth			

Write out these numbers in words:

$$10^3$$

$$10^{-6}$$

$$6 \times 10^1$$

$$3 \times 10^8$$

In this course you will sometimes need to multiply and divide numbers with exponents. When you multiply or divide numbers with exponents (they must have the same base number, usually 10) you simply add or subtract the exponents, respectively. If an exponent is raised to another power, like $(2^2)^2$, then the exponents are multiplied together. For example, $(2^2)^2 = 2^2 \times 2^2 = 2^4 = 2 \times 2 \times 2 \times 2 = 16$

A fractional exponent means to take the root of the number. The most common fractional exponent is $\frac{1}{2}$ which is the square root, so $16^{1/2} = \sqrt{16} = 4$. Similarly, the exponent $\frac{1}{4}$ is the fourth root, or the square root of the square root. So $16^{1/4} = \sqrt[4]{16} = 2$. Fractional exponents can be combined with integer exponents using the same rules described above.

Simplify the following problems involving exponents (indicate if it cannot be simplified):

$$10^6 \times 10^7 =$$

$$6^8 \times 6^{-5} =$$

$$10^{-4} \times 10^{-9} =$$

$$10^{-4} \times 10^8 =$$

$$6^8 \times 6^5 =$$

$$a^{-4} \times a^{-5} =$$

$$3^4 \times 4^3 =$$

$$2^3 \times 2^7 =$$

$$2^5 + 2^2 =$$

$$(3^2)^4 =$$

$$(4^3)^{1/2} =$$

$$(4^{3/2})^4 =$$

Scientific Notation

We can take advantage of exponents by using scientific notation to make extremely large and extremely small numbers easier to use. Numbers in scientific notation are expressed as a decimal between 1 and 10 followed by a power of 10, for example:

$$1.0 \times 10^8 = 100,000,000.$$

$$2.35 \times 10^4 = 23,500.$$

$$4.921 \times 10^{-3} = 0.004921$$

$$6.97 \times 10^{11} = 697,000,000,000.$$

Express these numbers in scientific notation:

725,000

0.03

64

• 0.00015

Multiplying numbers in scientific notation is simple- just multiply the leading numbers (called coefficients) and then add the exponents. If necessary, adjust the new coefficient to keep it between 1 and 10, and put the extra powers of 10 in the exponents.

For example:

$$(6.0 \times 10^4) \times (4.2 \times 10^5) = (6.0 \times 4.2) \times (10^4 \times 10^5) = 25.2 \times 10^9 = 2.52 \times 10^{10}$$

Try it yourself:

$$(1.2 \times 10^8) \times (3.0 \times 10^{-3}) =$$

$$(7.0 \times 10^4) \times (2.1 \times 10^{-2}) =$$

Applying Scientific Notation

Let's use scientific notation to compare some of the extremes in astronomy. The mass of one hydrogen atom is 1.67×10^{-27} kg, and the mass of the Sun is 2.0×10^{30} kg. If the Sun were pure hydrogen, how many hydrogen atoms would it contain?

The entire mass of the Milky Way Galaxy is about one and a half trillion times the mass of the Sun. Express this number in scientific notation and estimate the number of hydrogen atoms in the Galaxy.